The seed dispersal design lab

Teacher guide and worked answers

Explain how seed or fruit structures can aid dispersal and evaluate a controlled paper model.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsg/inputs/GROW SOW 2026-27.xlsx | GROW Weekly - Summer | C32

Investigate how seeds are dispersed.

New investigation outcome; dispersal models do not demonstrate evolutionary adaptation by need. Original secondary-age exemplar at an upper-KS2 conceptual level; no qualification approval or completion claim.

Prior learning

Know that seeds contain an embryo and can develop into plants.

Read repeated measurements in a table.

Success evidence

Link a structure to a dispersal mechanism.

Identify what changes and what stays the same in a model test.

Use repeat data without claiming measured outdoor dispersal distance.

Equipment

Shared seed/fruit cards and constructed test results, selected route, pencils; optional calculator.

Optional adult-only demonstration: two equal 4 × 16 cm paper strips; one folded into a compact shape, one with a simple rotor shape; metre rule and clear drop area.

Preparation

Default lesson uses the complete constructed dataset. Label any optional observations separately.

For a rotor: slit the top half lengthways into two equal blades; fold them in opposite directions. Fold the lower half into a narrow body. Keep all paper attached so mass is retained.

Use equal paper stock and original dimensions for both models; do not add clips or change the release height.

Practical care

No climbing, throwing or blowing seeds. If demonstrating, an adult releases models from a marked reachable height above a clear floor.

Use pre-cut paper where needed; no real berries, hooks or seeds are handled or tasted.

Ready to teach alternative

Use the supplied model design and constructed fall times. Pupils can investigate all decisions on paper without making or dropping anything.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	A seed has no engine
2	Retrieval	2-5	Seed, fruit and a fair comparison
3	I do	5-13	I connect structure to a journey
4	I do	Within stage	I design a fair paper comparison
5	I do	Within stage	I keep the claim within the measurement
6	We do	13-21	Model design jury
7	We do	Within stage	Detect a broken comparison
8	We do	Within stage	Choose the honest caption
9	Hinge	21-24	Which quantity was measured?
10	You do	24-34	Your independent investigation
11	You do	Within stage	Use the evidence
12	You do	Within stage	Check before sharing
13	Review	34-38	Compare and improve
14	Review	Within stage	A question worth investigating
15	Exit	38-40	Show the science you can explain

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening A seed has no engine

Invite a structure-based guess. Avoid intention language such as the seed wants to fly.

2 Retrieval Seed, fruit and a fair comparison

A new plant in suitable conditions; inside or associated with fruit structures depending on species, accept within fruit for core model; repeats reveal variation and strengthen confidence.

3 I do I connect structure to a journey

Think aloud: I name the structure before the mechanism. A wing supports interaction with air, not self-powered flight. Hooks support attachment, not growth towards an animal. Diagrams are selective.

4 I do I design a fair paper comparison

Shared I do time. Explain exact retained paper and optional rotor instructions from preparation. The constructed test uses no added clip. A clear drop avoids disturbance; pupils may remain entirely on paper.

5 I do I keep the claim within the measurement

Shared I do time. Think aloud: I cannot turn a fall-time result into a kilometre claim. This is a limited model of structure affecting descent, not proof of every dispersal outcome.

6 We do Model design jury

A wind; B external animal attachment; C animal ingestion. Rotor times exceed compact in every paired trial; use a pair such as 1.4 vs 0.8 seconds.

7 We do Detect a broken comparison

Shared We do time. No; mass also changed. Use the same retained paper mass and no extra clip, with matched release conditions.

8 We do Choose the honest caption

The rotor paper model took longer to fall in this setup; actual seed distance depends on wind, structure and environment. Model does not establish all-species ranking.

Reveal: In this paper test, the rotor fell more slowly. Real dispersal distance was not measured.

9 Hinge Which quantity was measured?

A is correct: the measured quantity is fall time. B is false: no outdoor distance was measured. C is false: the structure did not change intentionally.

A. The time each paper model took to reach the floor. - The units are seconds and the release conditions are stated.

B. How far every real seed travels outside. - The test used paper models indoors and did not measure distance.

C. How much the seeds wanted to move. - Wanting is not a scientific variable for a seed.

10 You do Your independent investigation

10 minutes across the three You do slides. Supply shared page 1 and one selected route page. All routes target the stated science objective; a labelled drawing or independent dictated explanation is valid.

11 You do Use the evidence

Shared You do time. Ask pupils which evidence supports their explanation. Read prompts or scribe without supplying the scientific conclusion.

12 You do Check before sharing

Shared You do time. Use the route-specific worked answers to identify one precise next step.

13 Review Compare and improve

4 minutes across Review. Offer partner or private teacher feedback. Ask for one specific revision linked to evidence; no public ranking.

14 Review A question worth investigating

Lundy cycle: invite written, spoken or pointed suggestions. Select one feasible question and explain how the pupil contribution changes the next example or investigation.

15 Exit Show the science you can explain

The wing interacts with air and slows descent. Accept an accurate pair of fall times. This paper model measures indoor fall time, not outdoor dispersal distance. E4 asks for one control.

Answers and assessment

R1-R3

New plant under suitable conditions; seeds are associated with fruit structures, core example inside fruit; repeats show variation and support a more reliable comparison.

W1-W3 / design fault

A wind; B external animal attachment; C animal ingestion. Rotor exceeds compact in all trial pairs: 1.4/0.8, 1.5/0.9, 1.3/0.7 seconds. Adding a clip changes mass as well as shape; match mass. W3 bounded paper fall-time conclusion, not all real seed distances.

Hinge A-C

A correct. B no distance measured. C intention is not the measured variable.

S1-S5

Air; animal fur; 1.4, 0.8, rotor; any stated control; no, paper indoor descent time is not an outdoor distance measurement.

N1-N5

Wing interacts with air/slows descent; hook attaches to fur; animals can ingest fruit and later pass surviving seeds. N2 correct paired values/units. N3 shape changes; mass, release height/rule and airflow controlled; time to floor measured. N4 added mass is another changed factor. N5 rotor took longer in this setup; limits include paper structure, indoor still air and no distance measurement.

T1-T4

T1 compact (0.8+0.9+0.7)/3=0.8 s; rotor (1.4+1.5+1.3)/3=1.4 s; difference 0.6 s. T2 longer descent may allow more wind influence, but actual distance depends on many conditions. T3 vary one shape under fixed airflow setting, release height and model mass; repeat and measure horizontal landing distance in cm. It is a planning task, not an instruction to improvise a fan practical now. T4 useful inherited structures can aid dispersal; individuals do not develop them because they want a ride. Evolutionary detail is optional.

E1-E4 / assessment

Correct structure-mechanism link; accurate pair and seconds; not real outdoor distance; relevant control/purpose. Secure: mechanism, fair variable distinction, numerical result and limit. Re-teach overclaim by underlining seconds and asking what was actually measured. Reflection accepts a precise next variable or measurement.

Teaching and assessment notes

40 minutes: opening 2; retrieval 3; I do 8; We do 8; hinge 3; You do 10; review 4; exit 2. Zero-minute slides share their stage time.

Supply shared page 1, one selected route page 2, 3 or 4, and page 5. Do not require all three routes. An independent spoken explanation can be recorded verbatim by an adult.

The default investigation is fully paper-based. Any optional adult demonstration is a separate observation, labelled as such; it must not overwrite the constructed sample record.

A sycamore dispersal unit is a winged fruit containing a seed; burdock burrs carry fruits/seeds. Keep the seed-versus-fruit distinction accurate without overloading the supported route.

Interactive decisions use the same structures, test conditions and times as W1-W3.

Access and responsive teaching

Supported

Match structure and mechanism; compare one pair of fall times.

Standard

Analyse repeats and explain controls and limitations.

Stretch

Calculate means and redesign a distance test while separating model from organism.

Regulation

Offer private writing, pointing, drawing or adult scribing. Allow pass-and-return for public questions and desk-based participation. Choose support from current learning evidence, not fixed ability labels.

Misconceptions to check

A wing means the seed flies using its own power.

Wind-dispersed structures interact with moving air; they do not choose or power a journey.

Check: Ask what moves the air or object.

Longer fall time proves a longer real outdoor journey.

It can give wind more time to act, but actual distance also depends on wind, height and other conditions.

Check: Ask what the experiment measured.

The plant grew a wing because it wanted to move.

Useful inherited structures exist through evolutionary processes, not an individual plant's intention.

Check: Ask whether a seed makes a decision.

Word	Meaning
dispersal	Movement of seeds away from where they formed.
fruit	A flowering-plant structure that develops from the ovary and contains seeds.
structure	The shape and arrangement of a part.
model	A simplified representation used to explore an idea.
control variable	A condition kept the same for a comparison.

Scientific references

Kew: Plant scientists classroom activity pack

Checked September 2026. Seed/fruit structures and dispersal examples. Scientific facts are paraphrased; this investigation is newly authored.

RHS: Seed dispersal

Checked September 2026. Winged and tufted structures support wind dispersal; a fall-time model does not measure outdoor distance.

Kew: How plants hitchhike on animal poo

Checked September 2026. Some animals eat fleshy fruit and later pass viable seeds; not every seed survives digestion.